

PlayWave Customer Churn Analysis

Turning 221 subscriber records into a revenue-at-risk model that tells Retention exactly who to save first — and why they're really leaving.

KEY FINDING

35.29% of customers have churned — and it isn't a support problem.

Escalations correlate with churn at just $r = 0.08$. The real drivers are price, content gaps, and trial-cancellation friction — while **\$1,715.22/month** in recurring revenue is already gone, and Standard-plan subscribers on Annual contracts are next.

221

Customers Analyzed

78

Churned

\$3,134.57

MRR (Active Users)

\$1,439.62

Avg. CLTV

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Tools: SQLite • SQL • Python (pandas) • Matplotlib • Seaborn

PlayWave is a fictional company created for this portfolio project. The dataset is synthetic; any resemblance to a real company is coincidental.

Business Context

For this self-directed project, I set out to solve a problem that comes up constantly in subscription businesses: understanding churn well enough to act on it. I modeled the scenario on PlayWave, a subscription streaming platform serving India and Nepal across Basic, Standard, and Premium tiers on Monthly and Annual contracts, acquiring subscribers through paid marketing, organic search, and referrals — the kind of business where cancellations fluctuate quarter to quarter with no single, unified view of who's leaving, why, or what it's costing.

Problem Statement

Without a consolidated view, retention campaigns, support staffing, and pricing decisions get made without evidence. I designed this project to answer, with numbers: which plans and geographies are highest-risk, how much monthly recurring revenue is exposed, and whether the top complaint drivers are actually about support quality — or about something support can't fix at all. This report walks through the end-to-end analysis I built to answer all three.

Data & Methodology

Data Sources (joined on customerid)

Customer Master	Demographics, country/state, gender, date of birth
Subscription	Plan type, contract type, monthly charges, CLTV, subscription/cancellation dates, cancellation reason, churn score
Customer Support	Complaint date, escalations, CSAT score

SQL

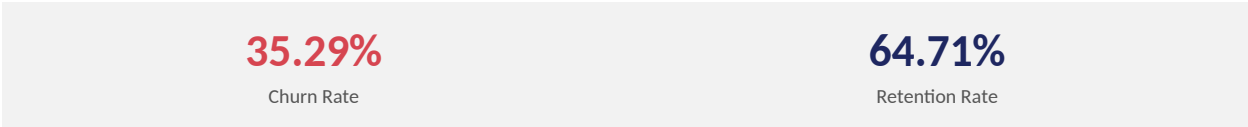
- Extracted all three tables from a SQLite database via SQL queries into pandas.

Python (pandas)

- Cleaned data: renamed columns, dropped irrelevant fields (interests, pincode), fixed date dtypes, standardized gender values, imputed missing country from state.
- Deduplicated support records — kept the latest complaint per customer, added a complaint_count feature.
- Joined all three sources into one analytical dataframe of 221 customers.
- Engineered churn_flag, tenure_days, and a Low / Medium / High churn_risk tier from churn_score (<50 / 50-70 / ≥70).
- Computed KPIs and segment breakdowns with groupby and pivot tables; visualized with Matplotlib and Seaborn (trend lines, bar charts, histograms, boxplots, a correlation heatmap, pairplot, and faceted catplot).

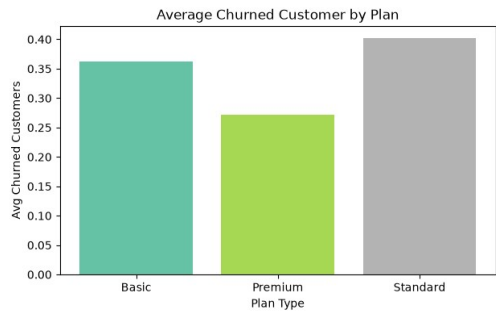
Key Findings

1. What share of customers churned?



So what: roughly 1 in 3 subscribers has cancelled — high enough that even small process fixes (like the trial-cancellation issue below) move real revenue.

2. Which plan and contract type churns most?

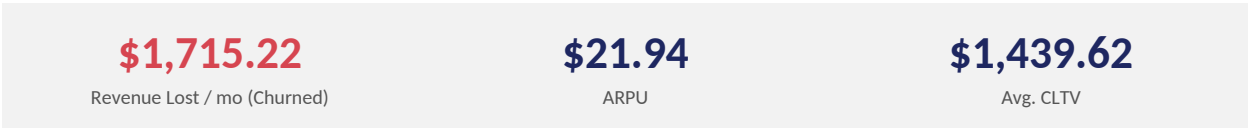


Standard 40.24% · Basic 36.25% · Premium 27.12%

Annual 37.38% vs. Monthly 33.33%

So what: Standard-plan, Annual-contract subscribers are the highest-value group to protect — enough to lose to feel it, enough left to make retention worthwhile.

3. How much revenue is at risk?

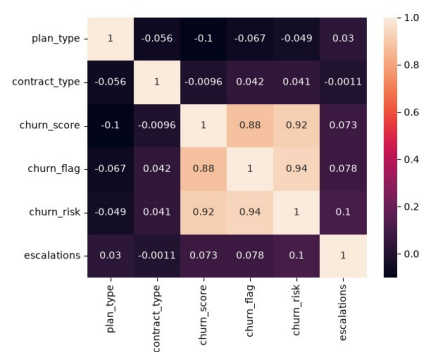


So what: at current ARPU, closing even a small share of the churn gap recovers meaningful MRR without new acquisition spend.

4. Why do customers say they're leaving?

- **Forgot to cancel trial — 18 cases** (a process fix, not a product problem)
- Too expensive — 17 cases
- Poor streaming quality — 16 cases
- Not enough content — 14 cases
- Switched to a competitor — 13 cases

5. Does support experience predict churn?



Escalation ↔ churn correlation: $r = 0.08$

Escalation rate 13.12% · Avg CSAT 49.46 · Avg complaints/user 0.46

So what: support quality is effectively unrelated to whether someone churns. The heatmap confirms churn_score ($r = 0.88-0.94$) — not support — is what actually tracks with cancellation.

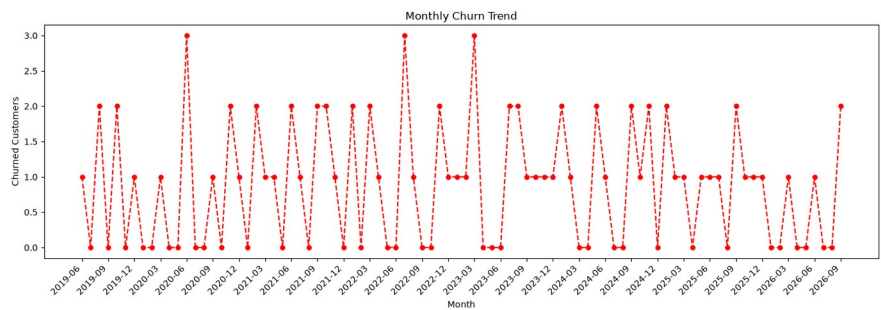
6. Which segments carry the most revenue risk?

Revenue at risk from monthly charges, by plan and churn-risk tier:

Plan	Total Charges at Risk	High-Risk Tier
Standard	\$1,891.18	\$685.73
Basic	\$1,843.20	\$514.75
Premium	\$1,115.41	\$217.84

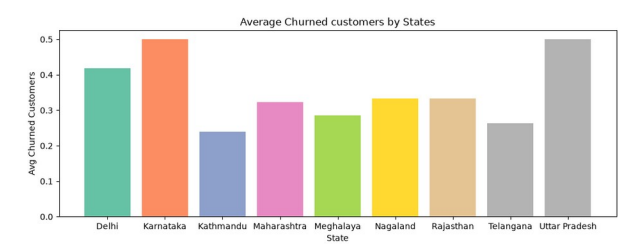
So what: Standard plan carries the single largest high-risk exposure (\$685.73) — it should be the first stop for proactive retention outreach.

7. How has churn trended over time?



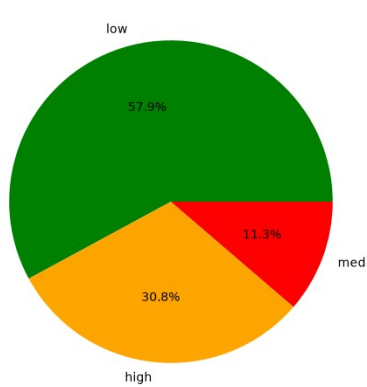
So what: churn is noisy month-to-month with no single dominant seasonal spike — consistent with churn being driven by ongoing price/content friction rather than a one-off event or campaign.

Where Churn Concentrates



Karnataka (50.00%) and Delhi (41.86%) churn highest; Kathmandu (24.00%) is the most stable market.

Customer Distribution by Churn Risk Tier



30.8% of customers already sit in the High-Risk tier — the pool this report prioritizes for outreach.

Acquisition Channel

Referral 38.81% · Paid 37.50% · Organic 30.49% — organically-acquired customers churn least, suggesting paid and referral channels may be bringing in less-committed subscribers or setting expectations the product doesn't meet.

Recommendations

1. Prioritize Standard-plan, Annual-contract retention outreach

This segment has the highest churn rate (40.24%) and the largest high-risk revenue exposure (\$685.73). Target them first with proactive check-ins ahead of renewal, since Annual customers give a longer runway to intervene before the cancellation decision is made.

2. Fix the trial-cancellation flow before touching support

"Forgot to cancel trial" is the single largest cancellation reason (18 cases) and is a pure process fix — add a reminder notification 2–3 days before trial-to-paid conversion and a one-click cancel path. This is the fastest, cheapest win in the whole dataset.

3. Investigate price and content, not support quality

With an escalation-churn correlation of just 0.08, support metrics are not the lever. "Too expensive" (17) and "Not enough content" (14) together outweigh support-adjacent complaints — route retention budget toward pricing/package tests and content-gap analysis instead of support headcount.

4. Build a High-Risk-tier save campaign, ranked by revenue

30.8% of customers are already High-Risk. Sequence outreach by the revenue table above — Standard, then Basic, then Premium — so retention effort is spent where it returns the most preserved MRR per contact.

5. Watch Karnataka and Delhi as leading indicators

These two states churn well above the 35.29% baseline (50.00% and 41.86%). A regional pilot of whichever fix (pricing, content, or trial flow) shows the biggest early effect there before rolling out nationally.

Limitations & Next Steps

- Sample size is 221 customers — segment-level rates (e.g., by state) are directional, not statistically definitive at this scale.
- Cancellation reason is self-reported at checkout and may understate price sensitivity relative to stated reasons like "content."
- Next: A/B test the trial-reminder fix and track its effect on the "forgot to cancel" bucket specifically; layer in a proper survival/CLTV model once more cohorts of data are available.

Skills Demonstrated

- SQL — multi-table extraction from SQLite
- Python / pandas — data cleaning, deduplication, feature engineering
- Exploratory data analysis & KPI computation (groupby, pivot tables)
- Customer segmentation (churn-risk tiering)
- Statistical correlation analysis
- Data visualization — Matplotlib & Seaborn (trend, distribution, correlation)
- Business storytelling — translating analysis into prioritized, revenue-linked recommendations